

C1MAT1T1 – O – 25 – 4064



FIRST SEMESTER B.SC. (REVISED NEP) DEGREE
EXAMINATION, JANUARY 2026
MATHEMATICS
Algebra – I and Calculus – I (DSC)

Time : 3 Hours]

[Max. Marks : 80

Instruction : Answer **all** questions.

I. Answer **all** questions :

(10×2=20)

- 1) State any two elementary row transformations.
- 2) State Cayley Hamilton theorem.
- 3) Discuss the continuity of the function $f(x) = \begin{cases} 1+x, & x \leq 2 \\ 5-x, & x > 2 \end{cases}$ at $x = 2$.
- 4) Evaluate $\lim_{x \rightarrow 0} \left(\frac{x - \tan x}{x^3} \right)$.
- 5) Find ϕ for the curve $r^2 = a^2 \cos 2\theta$ at $\theta = \frac{\pi}{6}$.
- 6) Define concavity and convexity of a curve $y = f(x)$.
- 7) When the curve $f(x, y) = 0$ is symmetric about the line $y = x$?
- 8) Find the n^{th} derivative of $y = e^{ax+b}$.
- 9) State Lagrange's mean value theorem.
- 10) Expand $\cos x$ in powers of x by Maclaurin's theorem.

II. Answer **any six** questions :

(6×5=30)

- 11) Find the rank of the matrix $\begin{pmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 5 \\ 1 & 2 & 3 & 4 \\ 1 & -2 & 6 & -7 \end{pmatrix}$.

- 12) For what values of λ and μ the system of equations
 $x + 2y + 3z = 5$
 $x + 3y - z = 4$
 $x + 2y + \lambda z = \mu$ have
a) No solution
b) Infinite solution
c) Unique solution ?

[P.T.O.]



15) If $\{a_n\}$ and $\{b_n\}$ are any two sequences, then prove that

$$\lim_{n \rightarrow \infty} (a_n b_n) = \lim_{n \rightarrow \infty} a_n \cdot \lim_{n \rightarrow \infty} b_n.$$

16) Prove that monotonic decreasing sequence which is bounded below converges to its infimum.

17) State and prove D'Alembert's ratio test for the convergence of series of positive terms.

18) Discuss the convergence of the series $\sum (\sqrt{n^2+1} - \sqrt{n^2-1})$.

III. Answer **any three** of the following :

(3×10=30)

19) a) Show that every subgroup of cyclic group is cyclic.

b) Show that the set Q_{-1} of all rational numbers except -1 with binary operation defined by $a*b = a+b+ab \forall a, b \in Q_{-1}$ is an abelian group.

20) a) Prove that the union of two countable set is countable.

b) State and prove Archimedian property of real numbers.

21) a) State and prove Cauchy's second theorem on limit of a sequence.

b) Prove that $\lim_{n \rightarrow \infty} \left(\frac{n^n}{n!} \right)^{\frac{1}{n}} = e$.

22) a) State and prove Leibnitz's test for alternating series.

b) Test the convergence of $\frac{x}{1.2} + \frac{x^2}{2.3} + \frac{x^3}{3.4} + \dots, x > 0$.



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MATHEMATICS

Algebra – I and Calculus – I (DSC)

Time : 3 Hours]

[Max. Marks : 80

Instruction : Answer all questions.

I. Answer all questions.

(10×2=20)

1) Define rank of matrix.

2) State Cayley-Hamilton theorem.

3) Discuss the continuity of the function.

$$f(x) = \begin{cases} 3x-2 & \text{if } x \leq 0 \\ x+1 & \text{if } x > 0 \end{cases}$$

at a point $x = 0$.

4) Evaluate $\lim_{x \rightarrow 0} \left(\frac{x - \sin x}{x^3} \right)$.

5) Define curvature of a curve at any point on it.

6) Find the asymptotes parallel to x and y -axes of the curve.

$$x^2y^2 - y^2 = a(x^2 + y^2)$$

7) When the curve $f(x, y) = 0$ is symmetric about x - axis ?

8) Find the n^{th} derivative of $\cos(ax + b)$.

9) State Rolle's theorem.

10) Expand e^x in powers of x by Maclaurin's theorem.

II. Answer any six questions.

(6×5=30)

11) Find the rank of the matrix,

$$\begin{bmatrix} 3 & -1 & -1 & 2 \\ 2 & 2 & 1 & -1 \\ 1 & -3 & 2 & -1 \\ 1 & -3 & 6 & -1 \end{bmatrix}$$

and hence reduce it into normal form.



12) For what values of λ and μ the system of equations.

$$x + y + z = 6$$

$$2x + 4y + 6z = 20$$

$$x + 2y + \lambda z = \mu$$

- a) a unique solution
b) Infinite solution
c) No solution.

13) If $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$ then prove that $\lim_{x \rightarrow a} [f(x) + g(x)] = l + m$.

14) Prove that every continuous function $f(x)$ in closed interval $[a, b]$ is bounded in that interval.

15) With usual notations prove that $\tan \phi = r \frac{d\theta}{dr}$.

16) Prove that the curves $r = a(1 - \cos \theta)$ and $r = b(1 + \cos \theta)$ cuts orthogonally.

17) State and prove Lagranges mean value theorem.

18) Find the n^{th} derivative of $e^{ax} \sin (bx + c)$.

III. Answer any three full questions.

(3×10=30)

19) a) Find the rank of the matrix,

$$\begin{bmatrix} -2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

and hence reduce into Echelon form.

b) Find the inverse of the square matrix $\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ using Cayley-Hamilton theorem.

20) a) State and prove intermediate value theorem.

b) Evaluate $\lim_{x \rightarrow 0} (\cos x)^{\frac{1}{x}}$.

21) a) Derive the formula for the radius of curvature of the curve in Cartesian form.

b) Find the pedal equation of the curve $2a = r(1 + \cos \theta)$.

22) a) State and prove Leibnitz theorem to find the n^{th} derivative of product of two functions.

b) If $y = (\sin^{-1} x)^2$ then prove that $(1 - x^2) y_{n+2} - (2n + 1)x y_{n+1} - n^2 y_n = 0$.